# 手掌关键点检测

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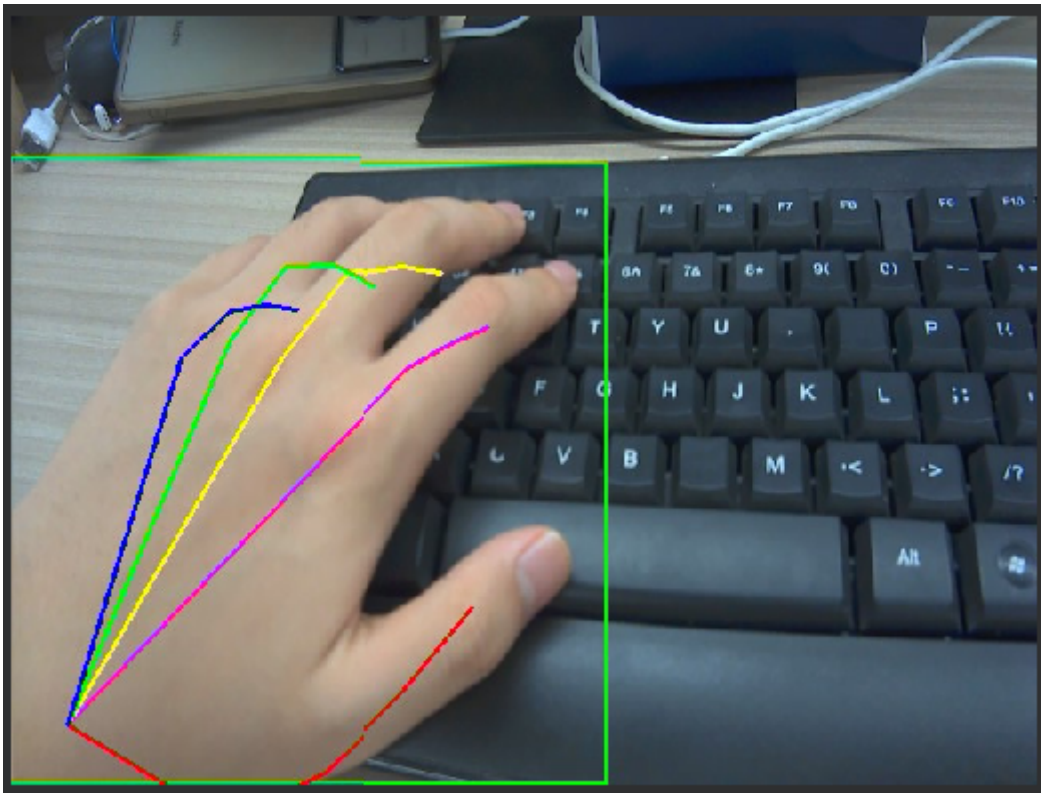
## 例程实验效果

本节我们来学习一下使用K230实现人类手掌的关键点的检测功能。

例程代码在【 源码汇总 / 08.Body / 06.hand_keypoint_detection.py 】

连接IDE后运行本节例程代码，用K230对准一张有人类的手存在的画面，可以看到屏幕中用不同颜色的线条标记出了手部的关键点的位置

> 由于该模型比较复杂，加载的过程花费时间较长，所以运行的前几秒只有画面，没有识别，这是正常情况



## 代码讲解

## 代码结构

**系统初始化/System Initialization:**

- 初始化参数设置/Parameter initialization
- 创建Pipeline/Create pipeline
- 初始化检测器/Initialize detector

**手掌检测流程/Hand Detection Process:**

- 图像预处理/Image preprocessing
- 手掌检测/Hand detection
- 检测框过滤/Detection box filtering

**关键点检测流程/Keypoint Detection Process:**

- 手部区域裁剪/Hand region cropping
- 关键点检测/Keypoint detection
- 坐标处理/Coordinate processing

**可视化显示/Visualization:**

- 绘制检测框/Draw detection box
- 绘制关键点/Draw keypoints
- 绘制连接线/Draw connections

**错误处理/Error Handling:**

- 异常捕获/Exception catching
- 资源清理/Resource cleanup
- 程序退出/Program exit

# 代码讲解

完整代码请参考【源码汇总 / 08.Body / 05.hand_keypoint_detection.py】文件

```
if __name__ == "__main__":
    # 显示模式，默认"hdmi",可以选择"hdmi"和"lcd" / Display mode, default is "hdmi",
can be "hdmi" or "lcd"
    display_mode = "lcd"
    rgb888p_size = [640, 360]

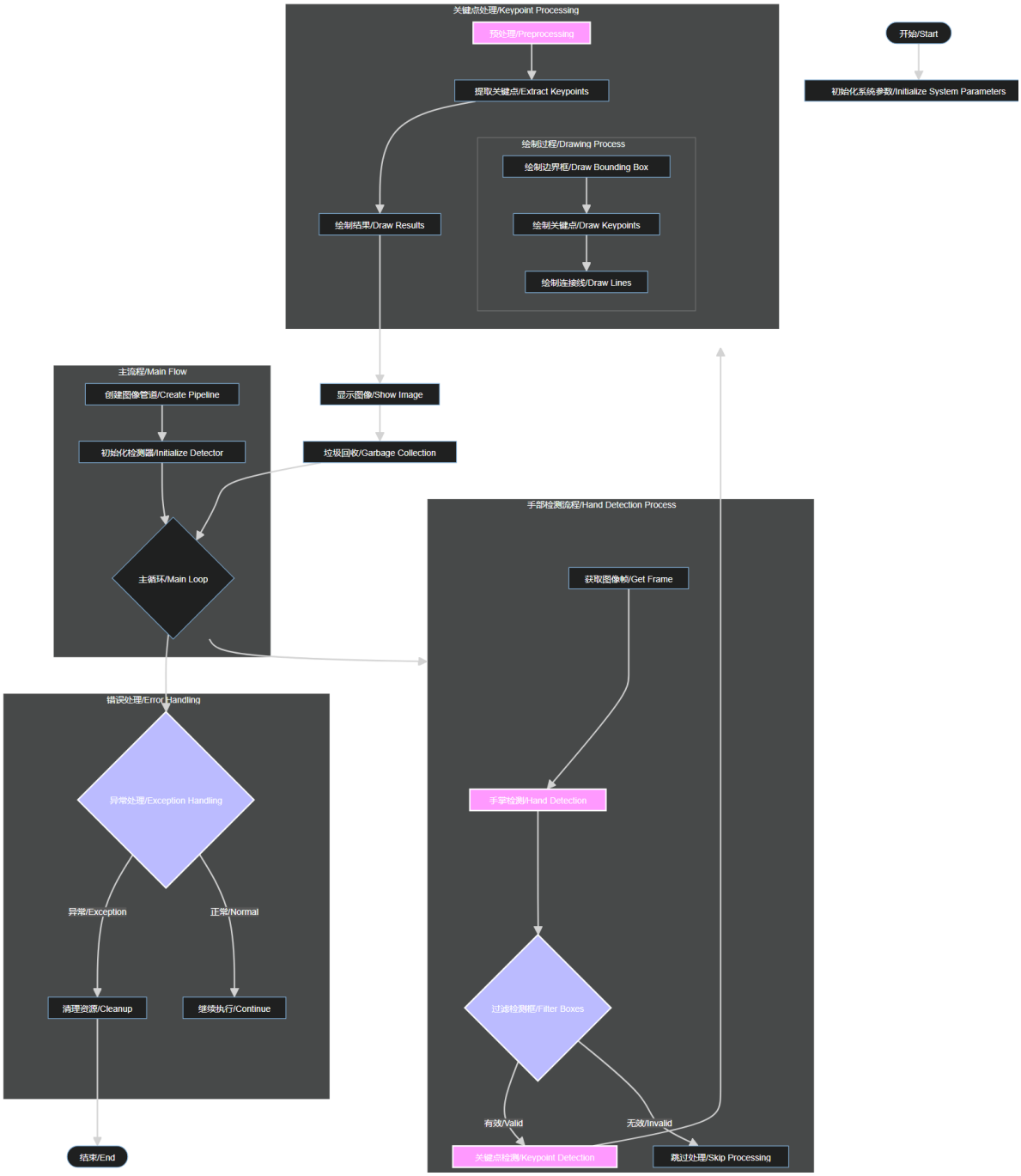
    # 根据显示模式设置显示尺寸 / Set display size based on display mode
    if display_mode == "hdmi":
        display_size = [1920, 1080]
    else:
        display_size = [640, 480]
    # 手掌检测模型路径 / Path to hand detection model
    hand_det_kmodel_path = "/sdcard/kmodel/hand_det.kmodel"
    # 手部关键点模型路径 / Path to hand keypoint model
    hand_kp_kmodel_path = "/sdcard/kmodel/handkp_det.kmodel"
    # 其它参数 / Other parameters
    anchors_path = "/sdcard/utils/prior_data_320.bin"
    hand_det_input_size = [512, 512]
    hand_kp_input_size = [256, 256]
    confidence_threshold = 0.2
    nms_threshold = 0.5
    labels = ["hand"]
```

```
    anchors = [26, 27, 53, 52, 75, 71, 80, 99, 106, 82, 99, 134, 140, 113, 161,
172, 245, 276]

    # 初始化PipeLine，只关注传给AI的图像分辨率，显示的分辨率
    # Initialize PipeLine, focusing only on the image resolution passed to AI and
the display resolution
    pl = PipeLine(rgb888p_size=rgb888p_size, display_size=display_size,
display_mode=display_mode)
    pl.create()
    # 初始化手掌关键点检测任务 / Initialize hand keypoint detection task
    hkd = HandKeyPointDet(hand_det_kmodel_path, hand_kp_kmodel_path,
det_input_size=hand_det_input_size, kp_input_size=hand_kp_input_size,
labels=labels, anchors=anchors, confidence_threshold=confidence_threshold,
nms_threshold=nms_threshold, nms_option=False, strides=[8, 16, 32],
rgb888p_size=rgb888p_size, display_size=display_size)

    # 主循环 / Main loop
    while True:
        with ScopedTiming("total", 1):
            img = pl.get_frame()                          # 获取当前帧 / Get current
frame
            det_boxes, hand_res = hkd.run(img)          # 推理当前帧 / Process
current frame
            hkd.draw_result(pl, det_boxes, hand_res)    # 绘制推理结果 / Draw
inference results
            pl.show_image()                               # 展示推理结果 / Show
inference results
            gc.collect()                                  # 执行垃圾回收 / Perform
garbage collection
            time.sleep_ms(5)                              # 短暂休眠减少CPU占用 /
Brief sleep to reduce CPU usage
    pl.destroy()
    hkd.hand_det.deinit()
    hkd.hand_kp.deinit()
```

## 流程图


关键点处理/Keypoint Processing
预处理/Preprocessing
提取关键点/Extract Keypoints
绘制过程/Drawing Process
绘制边界框/Draw Bounding Box
绘制关键点/Draw Keypoints
绘制连接线/Draw Lines
绘制结果/Draw Results
开始/Start
初始化系统参数/Initialize System Parameters
主流程/Main Flow
创建图像管道/Create Pipeline
初始化检测器/Initialize Detector
主循环/Main Loop
显示图像/Show Image
垃圾回收/Garbage Collection
手部检测流程/Hand Detection Process
获取图像帧/Get Frame
手掌检测/Hand Detection
过滤检测框/Filter Boxes
有效/Valid
无效/Invalid
关键点检测/Keypoint Detection
跳过处理/Skip Processing
错误处理/Error Handling
异常处理/Exception Handling
异常/Exception
正常/Normal
清理资源/Cleanup
继续执行/Continue
结束/End